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Remember: Any group members who did **not contribute to the project should be given all zero (0) points for the collaboration grade on the GWP submission page.*

Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).	
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Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed. **Note:** You may be required to provide proof of your outreach to non-contributing members upon request.

Step 1:

The portfolio consists of 5 stocks: Apple, Google, Microsoft, Amazon and NVIDIA.

The date range for training is between 2023-01-01 and 2023-06-31, and the back test is between 2023-08-01 and 2023-10-30.

The rate for 10-year treasury was used as a proxy for annualised risk-free rate, and it was 3.62% during the period used for training.

Step 2 and Step 3:

Below is a summary of the performance of different portfolio strategies from backtesting:

Portfolio Strategies	Cumulative Return (3M)	Volatility (Annualized)	Sharpe Ratio	Rank
Equal-Weight	59.69%	22.05%	1.6509	2
Markowitz	54.53%	23.81%	1.4347	3
Monte Carlo(1) - Max Sharpe Ratio	76.05%	25.98%	1.6729	1
Monte Carlo(2) - Max Sharpe Ratio	67.06%	24.23%	1.7393	1
Monte Carlo(3) - Max Sharpe Ratio	76.05%	25.98%	1.6596	1

It can be seen from the summary that all of the portfolio strategies performed well and achieved a cumulative return for more than 50%. Markowitz achieved the lowest return of 57.57% and lowest sharpe ratio of 1.55, and the equal-weight portfolio achieved a return of 59.69% and sharpe ratio of 1.65. Although the equal-weight strategy outperformed the Markowitz strategy, the difference is small. The best performing strategy was Monte Carlo that randomly selected weight for each security, showing the highest return of 75.21% and largest sharpe ratio of 1.73.

The better performance seen in the Monte Carlo portfolio can be explained by the increasing weight in the Nvidia and reducing weight in Apple.

	Equal-Weight	Markowitz	Monte Carlo (1)	Monte Carlo (2)	Monte Carlo (3)
AAPL	0.2000	0.7573	0.5409	0.5490	0.5409
MSFT	0.2000	0	0.0666	0.0500	0.0666
GOOG	0.2000	0	0.0133	0.0091	0.0133
AMZN	0.2000	0	0.0213	0.0901	0.0213
NVDA	0.2000	0.2427	0.3578	0.3019	0.3578

Step 4: Black-Litterman

2023 denoted an important year for the AI boom. Microsoft and Google were the two major players in the market, and NVIDIA was one of the leading manufacturers for AI chips.

4.1 Key Analyst Forecasts and Translated Black-Litterman Views:

1. TechCrunch - Microsoft invests billions more dollars in OpenAI, extends partnership (Wiggers, 2023)
    a. In Jan 2023
    b. Increasing investment means increasing expected growth
    c. View: Microsoft will outperform Google in the next quarter by around 6%.
2. Reuters - "Nvidia shares soar nearly 30% as sales forecast jumps and AI booms" (Reuters, 023)
    a. in May 2023
    b. Very strong positive news that Nvidia forecasted second-quarter revenue more than 50% above Wall Street estimates. Shares of Nvidia increased by 28% to its record high of $391.50.
    c. View: Nicidia will continue to grow for about 20% with AI bloom
3. Apple Newsroom - Apple reports second quarter results (Apple, 2023)
    a. In May 2023
    b. Mixed performance, with second quarter revenue of $94.8 billion down 3% year over year, and all-time high record for service revenue from App Store, Apple music etc.
    c. View: Apple will outperform Amazon (in service relatives revenue) for about 3%.

Confidence Matrix (Ω):

- High confidence in Nvidia and Microsoft growth and development.
- Medium-high confidence in Apple's performance as the second-quarter was of a mixed performance.

The analysis provides ready-to-use Black-Litterman inputs, including the P (picking) matrix, Q (view) vector, and Ω (confidence) matrix based on current institutional forecasts.

4.2 Comparison of back-testing performance

The Black-Litterman portfolio significantly out-performed the Markowitz and Equal-Weight portfolios with the highest sharp ratio of 1.65, and the largest return of 71.4%.

Portfolio Strategies	Cumulative Return (3M)	Volatility (Annualized)	Sharpe Ratio
Equal-Weight	59.69%	22.05%	1.6509
Markowitz	54.53%	23.81%	1.4347
Black-Litterman	71.40%	25.92%	1.6596

The excelent performance in Black-Litterman portfolio was due to the change in increasing the weight for Nvidia and Microsoft, while reducing the weight for Apple. Both portfolios did not hold any stocks for Google and Amazon. This could be due to the strong signals in the training dataset that Microsoft, Nvidia and Apple had stronger price increase and performance growth.

	Equal-Weight	Markowitz	Black-Litterman
AAPL	0.200	0.7573	0.6437
MSFT	0.200	0	0.0107
GOOG	0.200	0	0
AMZN	0.200	0	0
NVDA	0.200	0.2427	0.3456

4.3 Black-Litterman Model (or BL Model): Technical Background and Mathematical Framework

The Black-Litterman (BL) model, developed by Fischer Black and Robert Litterman at Goldman Sachs in 1990, addresses key limitations of traditional mean-variance optimization by incorporating investor views into portfolio allocation decisions. The model provides a Bayesian framework that combines market equilibrium assumptions with subjective investor opinions to generate more stable and intuitive portfolio weights.

The Core Mathematical Framework
The Heart of the Model

The Black-Litterman equation might look intimidating at first glance, but I've found it helps to think of it as a sophisticated weighted average. The central formula is:

μ_{BL} = [(τΣ)⁻¹ + P^TΩ⁻¹P]⁻¹[(τΣ)⁻¹π + P^TΩ⁻¹Q]

When someone explains this to someone, they usually tell them we're essentially asking: "Given what the market thinks (π) and what I think (Q), what should I actually expect?" The math gives us a principled way to blend these two sources of information.

Let us break down each component:

- **μ_{BL} is what we're solving for – our new expected returns**
- **τ reflects how much we trust the market's current pricing (Typically we use 0.025)**
- **Σ is our covariance matrix – how assets move together**
- **π represents what returns would justify current market weights**
- **P is how we specify which assets our views concern**
- **Q contains our actual return forecasts**
- **Ω captures how confident we are in our views**

Starting with Market Equilibrium

One insight that took time to appreciate fully is how we derive the equilibrium returns π. We're essentially reverse-engineering what the market "thinks" by asking: what expected returns would make the current market weights optimal?

π = λΣw_{mkt}

Here, λ represents the market's risk aversion (usually estimated at around 3), and w_{mkt} are the current market capitalization weights. This gives a sensible starting point that respects collective market wisdom.

The Updated Uncertainty

What one finds particularly elegant is how the model also updates our uncertainty about returns:

Σ_{BL} = [(τΣ)⁻¹ + P^TΩ⁻¹P]⁻¹

This isn't just academic – it directly affects how aggressively one will position his portfolio. More confident views lead to more concentrated positions.

Working with the Model in Practice

Specifying Views

Over the years, it has been noticed that view specification is more art than science. For absolute views like "US stocks will return 7% next year," one sets up the P matrix with a single 1 in the appropriate position. For relative views, such as "International stocks will outperform US stocks by 2%," I use [1, -1, 0, 0, ...].

The Q vector contains actual forecasts. What anyone would find crucial is being realistic about these numbers – they don't need to be dramatically different from equilibrium returns to have a meaningful portfolio impact.

Calibrating Confidence

The Ω matrix has given the most trouble over the years. How confident should one be? General consensus has been settled on a few practical approaches:

For high confidence views (based on strong fundamental analysis): ω = 0.01-0.05
For medium confidence views (based on moderate conviction): ω = 0.10-0.25
For low confidence views (hunches or weak signals): ω = 0.50 or higher

Parameter Selection

The τ parameter has always been somewhat controversial among practitioners they know. One has to experiment with different approaches:

- **The conservative 1/T approach (where T is sample size) often feels too restrictive**
- **Values around 0.025-0.05 seem to work well in practice**
- **Going much above 0.10 starts to overwhelm the equilibrium prior too much**

Step 5: Kelly Criterion

5.1 and 5.2

The Kelly Criterion portfolio did not out perform the Markowitz and Equal-Weight portfolios. It has the lowest sharp ratio of 1.45, and the lowest return of 52.54%.

Portfolio Strategies	Cumulative Return (3M)	Volatility (Annualized)	Sharpe Ratio
Equal-Weight	59.69%	22.05%	1.6509
Markowitz	54.53%	23.81%	1.4347
Kelly Criterion	52.54%	22.62%	1.4522

The worse performance in Kelly Criterion portfolio can be attributed to the change in reducing the weight for Nvidia and Apple. Both Markowitz and Kelly Criterion portfolios did not hold any stocks for Google and Amazon. This could be due to the strong signals in the training dataset that Microsoft, Nvidia and Apple had stronger price increase and performance growth.

	Equal-Weight	Markowitz	Kelly Criterion
AAPL	0.200	0.7573	0.6832
MSFT	0.200	0	0.1115
GOOG	0.200	0	0
AMZN	0.200	0	0
NVDA	0.200	0.2427	0.2053

5.3 The Kelly Criterion: Technical Background and Mathematical Framework

Introduction

The Kelly Criterion, developed by John L. Kelly Jr. at Bell Labs in 1956, solves a fundamental problem in finance and gambling: how much should you bet when you have an edge? This mathematical framework provides optimal position sizing that maximizes long-term wealth growth while managing downside risk by balancing the probability of success against potential payoffs.

The Core Mathematical Framework

The fundamental Kelly equation for a simple binary bet is deceptively simple:

$f = (bp - q) / b^*$

Let me break down each component:

- **f*** represents the optimal fraction of your total capital to wager
- **b** is the odds you receive (if you bet $1 and win, you get back $b)
- **p** is the probability of winning
- **q** is the probability of losing, which equals (1 - p)

What makes this formula so powerful is its output interpretation. When f* is positive, you have a favorable bet and should allocate that fraction of your capital. When f* is negative or zero, the bet is unfavorable and should be avoided entirely.

For example, consider a coin flip where you win $2 for every $1 bet, and the coin is biased in your favor with a 60% chance of heads. Here, b = 2, p = 0.6, and q = 0.4:

$f^* = (2 \times 0.6 - 0.4) / 2 = (1.2 - 0.4) / 2 = 0.4$

This tells us to bet 40% of our capital on each flip—a surprisingly aggressive recommendation that initially seems counterintuitive but maximizes long-term growth.

Extensions to Continuous Returns

In real financial markets, we rarely deal with simple binary outcomes. For investments with continuous return distributions, I've found the Kelly formula takes on a more familiar form:

$f = \mu / \sigma^{2*}$

Here, μ represents the expected excess return (return above the risk-free rate) and σ^2 is the variance of returns. This formulation reveals an important insight: the optimal allocation is proportional to expected return and inversely proportional to risk, measured by variance.

This continuous form connects Kelly betting to modern portfolio theory, showing that the growth-optimal portfolio has weights inversely related to the covariance structure of returns. For multiple assets, this extends to:

$\mathbf{f = \Sigma^{-1}\mu}$

where f is the vector of optimal portfolio weights, Σ^{-1} is the inverse covariance matrix, and μ is the vector of expected excess returns.

Parameter Estimation: The Practical Challenge

The theoretical elegance of Kelly's formula masks a significant practical challenge: accurately estimating the input parameters. In my experience, this is where most implementations succeed or fail.

Historical Estimation Approach

The most straightforward method uses historical data:

- Estimate p as the fraction of favorable outcomes in past data
- Calculate b from the average return ratio of winning trades
- Use sample variance for σ^2 in continuous applications

However, this approach assumes that past patterns will continue, which markets consistently violate. I've learned to treat historical estimates as starting points rather than gospel.

Bayesian Parameter Estimation

A more sophisticated approach incorporates prior beliefs with observed data. For probability estimation, we can use a Beta prior distribution:

If we start with prior beliefs Beta(α_0, β_0) and observe w wins and l losses, our posterior belief becomes Beta($\alpha_0 + w$, $\beta_0 + l$), giving us an expected probability of:

$$\mathbf{E[p] = (\alpha_0 + w) / (\alpha_0 + \beta_0 + w + l)}$$

This framework allows us to incorporate expert judgment or theoretical considerations alongside empirical evidence, which I find particularly valuable in domains with limited historical data.

Maximum Likelihood Methods

For continuous return distributions, maximum likelihood estimation provides another pathway. Assuming returns follow a normal distribution, we estimate μ and σ^2 by maximizing the likelihood of observed returns. While computationally intensive, this approach provides confidence intervals and goodness-of-fit measures that inform our parameter uncertainty.

Understanding the Assumptions and Limitations

Kelly's optimality comes with important assumptions that practitioners must understand:

1. **Independence**: Each bet or investment period is independent of others
2. **Perfect Information**: We know the true probabilities and payoffs
3. **Infinite Time Horizon**: We're maximizing long-term geometric growth
4. **Logarithmic Utility**: We care only about the growth rate, not the interim volatility
5. **No Leverage Constraints**: We can borrow or short-sell freely

Parameter Sensitivity Analysis

The Kelly fraction's sensitivity to probability estimates is particularly concerning. The partial derivative with respect to probability is:

$$\partial f/\partial p = (1 + b) / b^*$$

This shows that sensitivity increases with higher odds. A small overestimate of probability on a high-odds bet can lead to catastrophic over-betting. In my analysis, I've found that a 5% overestimate of win

probability can increase recommended bet size by 20-30%, highlighting the critical importance of conservative probability estimates.

Behavioral and Psychological Considerations

Beyond the mathematical framework, successful Kelly implementation requires understanding human psychology. The strategy can experience significant drawdowns that test investor resolve.

The path dependency of geometric growth means that large losses early in the strategy can permanently impair long-term performance, even if the underlying edge remains intact. This creates a tension between mathematical optimality and practical sustainability.

Conclusion

The Kelly Criterion provides a rigorous foundation for position sizing across applications from gambling to hedge fund management. Its geometric growth focus suits long-term investors with strong analytical conviction.

Successful implementation requires careful parameter estimation and often conservative modifications for uncertainty and psychological constraints. The Kelly Criterion works best as a conceptual framework rather than a rigid rule, forcing practitioners to quantify their beliefs and make more disciplined investment decisions.

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